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RV COLLEGE OF ENGINEERING®
 (An Autonomous Institution affiliated to VTU)
VII Semester B. E. Examinations Feb/Mar -2023
Computer Science and Engineering
COMPUTER GRAPHICS

Time: 03 Hours**Maximum Marks: 100****Instructions to candidates:**

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	What are the 3-primary subtractive colors and the 3-primary additive colors respectively?	01
	1.2	The parameter for glOrtho() function are _____.	01
	1.3	The color code "000" in RGB mode is for ____.	01
	1.4	In ___ projections, projectors are perpendicular to view plane.	01
	1.5	If the scaling factors values sx and sy<1 then it ___ the size of the object.	01
	1.6	What is the use of homogeneous coordinates and matrix representations?	01
	1.7	The point at which a set of projected parallel lines appear to converge in called as _____.	01
	1.8	A Bezier curve is a polynomial of degree _____ the number of control points used.	01
	1.9	A polygon in which the line segment joining any two points within the poygon lies completely inside the polygon is called____.	01
	1.10	What is a depth buffer?	01
	1.11	What is an event driven callback function?	02
	1.12	Differentiate Stroke text and Raster text.	02
	1.13	Differentiate between Flood fill and Boundary fill algorithm.	02
	1.14	List the drawback of DDA algorithm.	02
	1.15	State the different types of Multiview orthographic projection.	02

PART-B

2	a	Demostrate how RGB and indexed colour models in OpenGL is represented? Explain with a neat diagram.	05
	b	Discuss with a neat diagram, the graphics pipeline with major steps in the imaging process.	05
	c	With relevant examples discuss physical input device and Logical input devices.	06

3 a b	Explain Bresenham's line drawing algorithm to scan convert a line whose slope m is $0 < m < 1$. Scan convert a line using this algorithm for line connecting (5,8) to(9,11). Examine the working principle of Sutherland-Hodgeman polygon clipping algorithm for the polygon given in fig 3b against the clipping window. Hand-simulate all the intermediate processing steps and indicate the same with neat sketches. Fig 3b	08
4 a b	Write an OpenGL C program (using openGL fucntoins) to draw a circle with center (x_c, y_c) and radius r using Midpoint Circle algorithm. Provide mouse interface for the program to input the centre and radius of the circle. Examine Cohen Sutherland line clipping algorithm working principle for the following cases, considering suitable example (co-ordinates for endpoints): represent diagrammatically - Both endpoints of line inside the clipping window. - Both endpoints of line outside the clipping window. - Line intersecting one edge of clipping window. - Like intersecting two edges of clipping window.	08 08
5 a b	Write a C program to spin a cube using OpenGL transformation fiunctions. Give mouse interface. On clicking the left mouse button, it rotates about x-axis , on middle button it rotates about y-axis and on clicking right mouse button it rotates about z-axis. Explain rotaion translation and scaling with respect to 3-dimenstional coordinates.	08 08
6 a b	Discuss three-dimensional rotation of an object about an axis that is not parallel to one of the coordinates axis. Indicate all the steps involved with the aid of neat sketches. Compute the following 2D transformations: - Rotate a triangle A(0,0), B(2,2), C(4,2) about P(-2,-2) by an angle of 45 degree. - Illustrate the effect of applying reflection transformations on the square A(0,0), BB(1,0),CC(1,1) and D(0,1) with the aid of neat graphical representaions for the following cases: -about x-axis -about y-axis.	08 08

7	a	Write the complete taxonomy of classification of geometric projections with description.	06
	b	Give details of OpenGL functions related to projection.	06
	c	Mention the differences between parallel and perspective projections.	04
8	a	List the important properties of Bezier curves and <i>B</i> -spline curves.	06
	b	Briefly describe the visible surface problem and explain the binary space partitioning algorithm (BSP)	06
	c	Discuss the difference between object space and image space algorithm of hidden surface removal algorithms.	04